

Flow-Centred Design: Rethinking Accessibility as an Experiential Quality in Human-Computer Interaction

Daniel Hajas¹

Abstract. Accessibility standards and assistive technologies continue to improve social inclusion by enabling disabled people to access places, information, and services. However, accessibility in Human-Computer Interaction remains primarily evaluated through task completion, compliance, and usability, often overlooking the experiential quality of interaction once access has been achieved. This position paper proposes *flow* as a complementary lens for reframing accessibility beyond access. Drawing on flow theory from psychology, I argue that accessible systems should not only enable participation but also support conditions for *sustained engagement*, *guided exploration*, and *skill development*. To operationalise this notion, I introduce *Flow-Centred Design (FCD)* as a conceptual framework linking accessibility evaluation with interaction design practice. FCD reframes accessibility as an experiential quality of interaction, grounded in design principles derived from conditions associated with flow experiences. The framework also introduces measures for evaluating the experiential quality and extent of accessibility across different interaction contexts. While the discussion focuses primarily on the digital inclusion of blind people, the proposed framework may be applicable to broader forms of disability across both digital and physical accessibility contexts. Through examples from accessible data visualisation research, I illustrate how interaction techniques supporting customisation, structured exploration, perceptual congruence, and multimodal collaboration begin to move accessibility beyond task completion toward richer, more engaging, and growth-oriented experiences.

1 INTRODUCTION

Over the past several decades, disability rights advocates have fought to secure equal participation in education, employment, and public life [4, 1, 2, 17]. These civil rights movements, alongside technological innovation, have led to major advances in the development and adoption of accessibility standards, assistive technologies, and inclusive design practices. In digital environments, accessibility guidelines and compliance-based evaluation have become central mechanisms for ensuring that disabled people can access information and services. These developments represent an important and necessary foundation for digital inclusion.

However, accessibility is often framed in terms of whether a task can be completed at all. In practice, this framing can obscure an important question: *what is the quality of the experience once access is achieved?* For many blind users, interacting with digital systems involves navigating layers of friction [12]. This often takes the form of long, linear navigation through poorly structured interfaces, as well

as the need to plan and mentally simulate interaction steps in advance because systems behave unpredictably. Over time, the constant problem-solving and adaptation required to cope with these design shortcomings generates emotional and cognitive fatigue that makes it difficult to remain immersed in the activity itself. In other words, when interaction requires constant compensation for inaccessible design, part of the user's attention and energy is spent overcoming barriers rather than pursuing meaningful goals. Consequently, blind people may perceive digital interaction as a more negative experience than their sighted peers, even when they are able to complete the same task. Yet, accessibility lacks a clear conceptual vocabulary for specifying and evaluating the experiential quality of interaction once baseline access has been achieved.

In response, this paper proposes *flow* as a complementary lens for evaluating accessibility in Human-Computer Interaction. Typically, flow is described in terms of the *optimal experience*. According to Mihály Csíkszentmihályi [7], flow occurs when “one’s skills are adequate to cope with the challenges at hand, in a goal-directed, rule-bound action system that provides clear clues as to how one is performing.” Drawing on flow theory from positive psychology, I argue that accessibility should not only enable participation, but also support conditions for meaningful engagement through *sustained engagement*, *guided exploration*, and *skill development*. From this perspective, accessible systems are those that create conditions for flow experiences for blind people and, respectively, for other disabled individuals.

To make a tangible contribution, the paper introduces *Flow-Centred Design (FCD)* as a conceptual framework for specifying and evaluating the experiential quality of accessibility. FCD proposes a set of design principles associated with conditions for flow and introduces families of measures concerned with *customisability and adaptability*, *exploratory structure and navigation*, and *learning and transferable skill development*. Accessible data visualisation is then used as a case study to explore how these conditions and measures may be instantiated in practice. Shifting the attention from minimal access toward the experiential quality of interaction, I argue that evaluating accessibility through the lens of flow can support more engaging, confidence-building, and growth-oriented forms of participation.

1.1 Positionality statement

I write this paper from the perspective of a blind individual, who identifies with both research and practitioner roles, working at the intersection of digital accessibility, Human-Computer Interaction, and disability innovation. My professional practice includes commentary on inclusive design in action, development of assistive technology prototypes, and accessibility testing as a consultant. As a scholar,

¹ University College London, England, email: d.hajas@ucl.ac.uk

I engage in co-design with subject-matter experts to explore new interaction approaches in accessibility, and I rely on collaborative tools for the dissemination of research findings. While my perspective is shaped by lived experience navigating digital systems through touch and text-to-speech technology alone, I am far from being a power user of any particular assistive technology. In investigating flow in accessible interaction, insights from expert users — analogous to elite performers in other domains — may reveal patterns of practice and expertise that extend beyond my own experience. As such, I intend this paper as a contribution to an ongoing interdisciplinary conversation about what access is, and what it is not.

2 FROM ACCESS TO FLOW: SPECIFYING AND MEASURING THE EXTENT OF ACCESSIBILITY AS AN EXPERIENTIAL QUALITY

In this section, I argue that accessibility should be reconsidered through the lens of *flow* — an optimal experiential state characterised by focused engagement, perceived control, and intrinsic reward[6]. Describing and building on characteristics of flow from existing literature in positive psychology, and their growing relevance within Human-Computer Interaction (HCI), I introduce *Flow-Centred Design (FCD)* as a framework for conceptualising accessibility as an experiential quality rather than solely a condition of access. Through FCD, this section pursues two aims: first, to specify the experiential quality of accessibility through a set of design principles derived from characteristics of flow experiences; and second, to propose directions for measuring the extent of accessibility through evaluative measures aligned with these principles.

2.1 Access is not enough

Research and practice in digital accessibility have traditionally focused on removing barriers to interaction between disabled people and digital content. Accessibility evaluation tools and frameworks typically ask whether systems comply with guidelines, and whether users can complete tasks, perceive content, or navigate interface elements using assistive technologies[22]. These assessment criteria are essential for establishing conditions for baseline inclusion. However, they primarily measure possibility rather than experiential quality and neglect aspects of experience design[10] and developmental flourishing[20]. As a result, a system may technically satisfy accessibility requirements while still imposing significant friction in interaction; requiring sustained cognitive effort, fragmented navigation strategies, and external tools for gaining proficiency.

Consider Silvia, a blind professional using a workplace chat application that is technically compatible with a screen reader and therefore considered accessible under conventional criteria. They can send messages, review conversations, and navigate channels independently without assistance. However, the interaction may still be effortful, fragmented, and cognitively demanding in ways that diminish sustained use. For example, when reviewing messages sent earlier in a conversation, screen reader focus may repeatedly jump back to the message input field whenever a new message arrives, interrupting concentration and forcing the user to reconstruct conversational context. In threaded replies, the interface may announce quoted messages before the new response itself, requiring additional effort to infer what information is novel. Similarly, metadata such as the sender's name may be repeatedly announced in one-to-one conversations where identity is already known, increasing verbosity and

interaction fatigue. Although such an application may satisfy accessibility requirements in the sense that interaction is possible, the experience itself may remain frustrating, tiring, and difficult to sustain. Navigation becomes fragmented, contextual awareness is repeatedly disrupted, and users may seek to complete interactions as quickly as possible rather than remain meaningfully engaged. In this sense, accessibility alone does not guarantee an interaction environment conducive to focused, confident, or rewarding participation.

2.2 Accessibility beyond access

I argue that accessibility evaluation should extend beyond barrier removal, to consider whether interaction environments support conditions for the state of flow. When an interaction is accessible and the optimal experience is achieved — that is, the user is in flow — users can direct their attention toward pursuing clear goals. In doing so, they can displace self-consciousness with conscious effort, feel and report positive emotions, and become more proficient, without the distractions of managing the interface itself. However, flow experience is inherently subjective and context-dependent. It arises from the dynamic relationship between a person's skills, goals, and the challenges presented by a task [6].

Although flow experiences are difficult to design for directly, prior work in positive psychology identifies a set of recurring characteristics associated with entering and sustaining flow states[6, 19]. These characteristics describe the experiential conditions under which individuals become deeply absorbed in meaningful activity, balancing challenge, attention, feedback, and perceived control. While originally formulated in domains such as play, sport, work, and creative practice, these conditions offer a useful conceptual foundation for reconsidering accessibility as an experiential quality of interaction. Several conditions are particularly relevant to accessibility contexts[19]:

1. Perceived challenges and opportunities for action should stretch skills and abilities, though not excessively.
2. Goals should be relatively short-term (proximal), specific, and detailed.
3. Immediate and actionable feedback should be available to promote continuing progress.
4. Concentration should be focused and intense.
5. Action and awareness should merge.
6. Reflective self-consciousness should fade or be lost.
7. A sense of control over actions and responses to changing situations should increase.
8. Time may become distorted, often appearing to pass more quickly than in reality.
9. Experiences should feel intrinsically rewarding, such that the activity itself justifies the effort involved.

In accessibility contexts, this variability of flow experiences is particularly pronounced. Blind users employ diverse interaction strategies and assistive technologies, often shaped by individual workflows and long-term adaptation to specific tools [21]. To operationalise flow as a measure of accessibility, I propose *Flow-Centred Design (FCD)* as a conceptual framework that links accessibility evaluation with experience design practice. Rather than prescribing a single optimal interaction technique, FCD treats flow as an evaluative heuristic — a way of identifying where friction disrupts engagement, where interaction demands exceed available skills, and where opportunities exist to support more perceptually congruent interaction [14]. In this

framework, accessibility is reframed not only as a condition of access, but as an experiential quality that may be supported to varying extents across different systems, users, and contexts.

Through FCD, two related aims are pursued. First, the experiential quality of accessibility is specified through a set of design principles derived from characteristics of flow experiences and adapted to accessibility contexts. Second, the extent of accessibility is considered through possible measures aligned with these principles, offering an initial conceptual vocabulary for evaluating how accessible interactions are experienced in practice.

2.3 Design principles for specifying the experiential quality of accessibility

The nine conditions outlined above describe characteristics associated with entering and sustaining flow experiences, but they are not immediately actionable as design goals for accessibility. To operationalise these characteristics in accessibility contexts, I synthesise and transpose them into three higher-level design principles that reflect recurring experiential qualities of interaction. Rather than reproducing the original psychological constructs directly, these principles adapt conditions of flow to concerns central to accessible interaction, such as reducing friction, maintaining orientation, supporting confidence, and enabling progressive growth in competence. Mapping the conditions for flow experiences to these principles, I propose that accessible systems should support *sustained engagement*, *guided exploration*, and *skill development* in meaningful activities, including learning, collaborative work, and creative expression.

Sustained engagement: Accessible systems should support the conditions necessary for maintaining focused and uninterrupted participation in meaningful activity. This principle aligns most strongly with conditions related to concentration, immersion, and intrinsic motivation. Specifically, flow requires *focused and intense concentration* (4), the *merging of action and awareness* (5), and the *fading of reflective self-consciousness* (6), all of which depend on interactions becoming sufficiently effortless that users can attend to the activity rather than the interface itself. In accessibility contexts, excessive friction — such as fragmented navigation, high cognitive load, or inaccessible controls — redirects attention toward managing the interaction, disrupting engagement. Sustained engagement also relates to *intrinsic reward* (9), whereby interaction feels meaningful and worthwhile beyond task completion alone. Flow-centred accessible systems should therefore minimise avoidable attentional disruption and support prolonged, rewarding engagement in activities such as reading, communication, creative work, or learning.

Guided exploration: Accessible systems should provide structure and feedback that support confident navigation through unfamiliar or evolving interaction spaces. This principle maps most closely to conditions concerning *proximal, specific goals* (2), *immediate and actionable feedback* (3), and an increased *sense of control* (7). Flow emerges when users can orient themselves toward clear objectives, understand the consequences of their actions, and adapt to changing circumstances without uncertainty or disorientation. In accessibility contexts, guided exploration may involve clear navigation pathways, progressive disclosure of complexity, predictable interface behaviour, and timely multimodal feedback that supports orientation and decision-making. Rather than prescribing a single interaction strategy, systems should enable users to explore while preserving confidence that progress is understand-

able, reversible, and responsive to individual workflows and assistive technologies.

Skill development: Accessible systems should support gradual increases in competence by balancing challenge with the user's evolving abilities. This principle aligns primarily with the requirement that *challenges and opportunities for action stretch skills without exceeding them* (1), creating conditions for growth rather than frustration or boredom. Over time, interactions should scaffold progression from effortful engagement toward fluency, allowing users to build proficiency with tools, workflows, and interaction strategies. Skill development also relates to the altered *experience of time* (8), often associated with deep engagement in progressively mastered activities, where sustained practice feels absorbing rather than effortful. In accessibility contexts, systems should support learning through adaptable levels of complexity, opportunities for mastery, and interactions that evolve alongside user expertise, enabling long-term flourishing rather than merely short-term access.

To illustrate these principles in practice, consider *Sophie*, a blind professional completing her weekly grocery shopping using a screen reader-accessible mobile application. Her primary cognitive effort should be directed toward meaningful decisions: determining how much she can afford to spend that week, estimating what can fit in her refrigerator, planning meals before ingredients expire, and reasoning about which items are needed for breakfast, lunch, and dinner. A flow-centred accessible system would support *sustained engagement* by minimising friction in interaction, allowing Sophie to compare products through a clear list structure, quickly add items to her basket without losing browsing context, and remain focused on meal planning rather than interface management. *Guided exploration* would be supported through predictable navigation, actionable feedback, and clear system state awareness; for example, immediate confirmation when products are added to the basket, notifications of out-of-stock substitutions, updates on spending totals, and reminders of the number of items selected. Over time, *skill development* would emerge as the system adapts to Sophie's preferences, enabling her to shop more efficiently, discover cost-saving alternatives, reuse prior shopping patterns, and gradually optimise her decisions around saved favourites and meals. In this example, accessibility is not only reflected in whether shopping is technically possible, but in the extent to which the interaction enables focused, confident, and rewarding participation in an everyday activity.

2.4 Measuring the extent of accessibility

If accessibility is understood as an experiential quality, then accessibility evaluation must also move beyond determining whether interaction is technically possible. The second contribution of Flow-Centred Design is therefore to suggest how the *extent of accessibility* might be evaluated in relation to the design principles introduced above: *sustained engagement*, *guided exploration*, and *skill development*. Rather than treating these principles as abstract aspirations, FCD proposes that they may serve as evaluative lenses through which the experiential quality of accessibility can be examined. This does not imply that flow can be measured simply or universally. Indeed, prior work identifies substantial variation in how flow is measured, with no single instrument adequately determining whether a person is in or out of flow [16]. For this reason, the measures proposed below should be understood as provisional, non-exhaustive families of evaluation for assessing how accessible interactions are experienced in practice.

Sustained engagement (through interaction friction and adaptation):ceptual vocabulary for examining the extent to which interactions support *sustained engagement*, *guided exploration*, and *skill development*. By reframing accessibility as an experiential quality, Flow-Centred Design offers a way of identifying where interaction supports or disrupts opportunities for meaningful participation and developmental flourishing. In the following section, accessible data visualisation is used as a case study to examine how these design principles and associated measures may be applied in practice, demonstrating both the strengths and limitations of contemporary interaction techniques.

The extent of accessibility may be evaluated by examining the degree to which interactions reduce unnecessary friction and adapt to users' abilities, preferences, goals, and evolving expertise. In Flow-Centred Design, accessible systems should support sustained engagement by maintaining an appropriate balance between challenge and skill, recognising that interaction strategies vary considerably across users and contexts. Measures may therefore include the *degree of customisability*, such as what content is spoken, how verbose this content is, or when it is read in sequence in the context of screen reader interaction[11]; the *extent of personalisation*, including adaptive support for different expertise levels, mediated by semantic scaffolding[25]; the *degree of user control*, reflected in whether interactions can be modified or tailored according to individual workflows; as well as the *extent of interactional friction*, reflected in interruptions to concentration, repeated corrective actions, or reliance on compensatory strategies. Rather than assuming a single optimal interaction pattern, greater accessibility may be reflected in systems that support multiple ways of engaging with information while reducing avoidable effort and adapting interaction demands over time.

Guided exploration (through orientation and navigational support):

The extent of accessibility may also be reflected in how effectively systems support orientation and confident exploration through interaction. Accessible interactions should provide meaningful structures that help users orient themselves, navigate confidently, and progressively deepen engagement with information. Measures may include the *mode of interaction structure*, such as whether information is organised through lists, tables, trees, graphs, or other representations suited to different tasks and reasoning strategies; the *diversity of navigational pathways*, including sequential, structural, spatial, or targeted navigation methods[23]; as well as the fluidity, reversibility, predictability, and contextual continuity of movement between states[14]. Systems that reduce disorientation, preserve context, and support independent exploration may therefore be perceived as more accessible than those requiring fragmented or effortful navigation.

Skill development (through progressive competence and transfer):

Finally, the extent of accessibility may be evaluated through a system's ability to support increasing competence and transferable proficiency over time. Flow-Centred Design proposes that accessible systems should not only enable participation in the present moment, but also facilitate progression toward greater competence, confidence, and autonomy. Measures may include the *rate of skill acquisition*, reflected in reductions in support needs — for instance, observed through less frequent prompting of a large language model-based Q&A assistive system, reduced navigation effort, or improved task completion time; the *transferability of interaction patterns*, where skills learned in one context can be applied across representations or tasks[14, 24]; and the *development of user confidence and independence*, reflected through qualitative feedback or longitudinal engagement. From this perspective, accessibility extends beyond immediate usability to encompass whether systems help users become increasingly capable participants in meaningful activity over time.

The measures proposed here are intentionally provisional and should be understood as illustrative families of evaluation rather than settled or comprehensive metrics. Their purpose is not to prescribe universal indicators of accessibility, but to provide an initial con-

3 ACCESSIBLE DATA VISUALISATION AS A CASE STUDY

Accessible data visualisation provides a useful case study for examining how interaction design can move beyond basic accessibility toward more experiential forms of engagement. In recent years, research in accessible data visualisation has increasingly shifted from ensuring access to graphical information toward enabling blind and partially sighted users to actively explore, manipulate, and reason about data through alternative representations. This emerging body of work provides a useful lens for examining how the design principles and measures proposed by Flow-Centred Design may be instantiated in practice. The following examples are not presented as definitive validations of FCD, but as demonstrative cases that illustrate how the experiential quality of accessibility may be supported and evaluated across different forms of interaction.

3.1 Structured exploration through accessible navigation

Rich Screen Reader Experiences for Accessible Data Visualisation [23] proposes a structured approach to screen reader interaction that moves beyond static textual descriptions of charts. Rather than presenting data visualisations as linear summaries, the system organises information hierarchically, enabling users to navigate between overview and detail, inspect subsets of data, and pursue analytical questions through structured exploration. This interaction model transforms screen reader use from passive information consumption toward active reasoning, allowing users to query and compare information according to their goals rather than relying on fixed descriptions. In doing so, the work reframes accessible visualisation as an exploratory process in which users can progressively deepen their understanding of a dataset.

From the perspective of Flow-Centred Design, this work aligns most strongly with the principle of *guided exploration*. Rather than overwhelming users with a single monolithic description, the hierarchical organisation supports gradual orientation and movement through the information space. Users are able to establish proximal goals, receive immediate feedback through interaction, and maintain a sense of control over their exploration, corresponding to several conditions associated with flow. The system also indirectly supports *sustained engagement*, as reducing navigational friction enables attention to remain directed toward reasoning about data rather than managing the interface itself. At the same time, reliance on hierarchical traversal may introduce limitations for some users. If users' mental models do not align with the imposed tree structure, or if interaction conventions differ from established screen reader workflows, exploration may become fragmented or cognitively effortful. For example, a user of Olli expected the tree structure to expand horizontally, similarly to a directory tree in Windows File Explorer, and

not vertically as it is currently implemented[3]. This suggests a possible tension between providing structured guidance and preserving low-friction, user-led exploration.

This work also helps contextualise how the extent of accessibility may be evaluated through the proposed family of evaluation related to *orientation and navigational support*. First, the system demonstrates one possible *mode of interaction structure*: a tree hierarchy that organises information into progressively deeper levels of detail. While this structure supports overview-to-detail exploration, later work demonstrates that alternative structures, such as graphs[8] or hypergraphs[14], may support different forms of reasoning. Second, the system expands the *diversity of navigational pathways* available to users, supporting structured traversal, spatial movement, and targeted search between levels of abstraction. From an FCD point of view, accessibility may therefore be understood not only in terms of whether a visualisation can be accessed, but in how richly users can navigate, orient themselves, and independently explore information.

3.2 Adaptable interaction through customisation

“Customization is Key”: *Reconfigurable Content Tokens for Accessible Data Visualisations* [11] extends accessible screen reader interaction by enabling users to actively reconfigure how information is presented. Rather than assuming a fixed description format, the system introduces configurable content tokens that can be permanently reordered, temporarily prioritised, and toggled or adjusted according to user preference and analytical goals. Thus, users are able to modify verbosity levels, foreground information relevant to a particular task, and tailor descriptions to suit both novice and expert workflows. In doing so, the work highlights a central challenge in accessibility: interactions that are effective for one user, context, or level of expertise may impose unnecessary friction for another.

This work aligns closely with the principles of *sustained engagement* and *guided exploration*, while also supporting *skill development* over time. By enabling users to adapt the interaction according to their preferences and expertise, the system helps maintain an appropriate balance between challenge and skill — a core condition of flow. For novice users, richer descriptions and greater guidance may reduce uncertainty and support orientation, whereas experienced users may benefit from concise, lower-friction interactions that enable faster reasoning. The ability to progressively modify interaction density also supports long-term proficiency, allowing workflows to evolve as users gain confidence and expertise. However, the act of customisation may itself introduce friction. Configuring verbosity, prioritising content tokens, or repeatedly adjusting interaction settings may interrupt concentration and temporarily disrupt sustained engagement. This highlights a potential tradeoff in Flow-Centred Design between adaptability and interactional continuity, particularly when configuration overhead outweighs experiential benefits.

Evaluating the accessibility, one could argue that the interface is fully accessible — without further discussion, since the fundamental interaction builds on the Olli interface[3]. However, this work strongly illustrates how the extent of accessibility may fall on an experiential spectrum, differentiated through the family of measures related to *interaction friction and adaptation*. Suitable measures may include the *degree of customisability*, reflected in the number and variety of ways content tokens can be reordered or configured; the *degree of user control*, reflected in the ability to modify verbosity, prioritise information, or adjust interaction pathways; and the *extent of personalisation*, reflected in how effectively the system accommodates differing expertise levels and goals. The work also reinforces

the relationship between adaptation and guided exploration, as adjustable structures and verbosity settings can influence how confidently and independently users navigate information spaces. Accessibility may therefore be understood not only as access to content, but as the extent to which interaction can be adapted to sustain meaningful and efficient engagement.

3.3 Transferable interaction through perceptual congruence

The concept of *perceptual congruence*, introduced through *Benthic: Perceptually Congruent Structures for Accessible Charts and Diagrams* [14], extends accessible visualisation by focusing not only on whether information can be accessed, but on how interaction structures correspond to the organisational logic of graphical representations. Rather than flattening charts and diagrams into linear descriptions or relying solely on hierarchical structures, Benthic introduces a hypergraph-based interaction model that preserves both hierarchical and relational groupings. This approach enables blind users to traverse representations fluidly, move between overlapping contexts, and reorganise exploration according to analytical goals. In doing so, the system supports forms of navigation and reasoning that more closely parallel the flexibility of sighted interaction, allowing users to develop richer mental models of diagrams and visualisations.

Here, perceptual congruence aligns strongly with the principles of *guided exploration* and *skill development*. The system supports guided exploration by enabling users to navigate complex information spaces in ways that are predictable, reversible, and responsive to changing goals. Rather than following fixed traversal paths, users are able to flexibly reorganise exploration, compare alternative perspectives, and recover from navigational detours without losing context. At the same time, Benthic supports transferable skill development through consistency in interaction logic across different graphical representations. By preserving underlying structural relationships, users can increasingly rely on learned reasoning strategies rather than repeatedly adapting to new interaction conventions, allowing skill acquisition to accumulate over time. Nevertheless, richer interaction structures may also increase cognitive demands, particularly for novice users unfamiliar with graph-based navigation or overlapping relational contexts. While perceptual congruence may support transferable reasoning strategies over time, early interactions may require greater effort to establish orientation and develop effective mental models.

Benthic illustrates how the extent of accessibility may be contextualised through both *orientation and navigational support* and *progressive competence and transfer* measures. First, the work expands the possible *mode of interaction structure* beyond tree hierarchies toward graph and hypergraph representations, demonstrating that accessibility may depend not only on whether structure is present, but on how appropriately that structure supports reasoning about relationships. Second, the system broadens the *diversity of navigational pathways*, enabling lateral, hierarchical, and reversible traversal, as well as flexible regrouping of information around user goals. Finally, perceptual congruence introduces a basis for evaluating the *transferability of interaction patterns*, reflected in whether users can apply familiar navigation strategies and mental models across domains such as charts, diagrams, or multimodal representations.

3.4 Multimodal and adaptive interaction ecologies

Umwelt: Accessible Structured Editing of Multimodal Data Representations [24] proposes a multimodal environment for creating and interacting with data representations in which visual, textual, and auditory modalities are treated as co-equal rather than hierarchical outputs. Instead of positioning accessible representations as secondary translations of visual artefacts, Umwelt derives multiple representations from a shared underlying data model, allowing users to fluidly move between modalities according to task demands and personal preferences. Users are able to independently author, modify, and reconfigure multimodal analytical workflows, switching between overview and detail while preserving context across representations. Consequently, Umwelt reframes accessibility as an ecosystem of interaction possibilities rather than a single optimised interface.

Umwelt represents perhaps the clearest illustration of how the three design principles may converge in practice. The system supports *guided exploration* through synchronised multimodal representations that preserve orientation and enable users to move flexibly between different ways of reasoning about the same information. It supports *sustained engagement* by allowing users to focus on analytical goals rather than on overcoming modality-specific barriers, reducing friction through complementary interaction pathways. Finally, Umwelt promotes learning and transferable *skill development* by enabling users to progressively refine workflows, build confidence across modalities, and adapt interaction strategies according to changing expertise or goals. Rather than prescribing a single ideal interaction, the system enables a dynamic balance between challenge and skill through user-led adaptation. At the same time, the availability of multiple modalities and interaction pathways may increase complexity and decision-making demands. Without sufficient guidance or Prior experience, users may exhibit uncertainty regarding which modality or workflow best supports a given analytical goal — a research topic often discussed, but not formally established.

Umwelt demonstrates how all three proposed families of evaluation may be contextualised within a single interaction environment. In terms of *interaction friction and adaptation*, the system supports extensive user control through runtime configuration, modality selection, and the ability to tailor representations according to analytical intent. Second, in case of *orientation and navigational support*, accessibility may be reflected in the diversity of available pathways, including movement between textual hierarchies, sonification, and visual representations while maintaining shared context. Finally, regarding *progressive competence and transfer*, Umwelt provides a basis for evaluating how users develop increasingly effective workflows, transfer reasoning strategies between modalities, and achieve greater independence in analytical tasks over time.

These case study examples suggest that accessible interaction design is increasingly moving beyond enabling task completion toward supporting richer forms of engagement. Across these systems, accessibility is not treated merely as the removal of barriers, but as the creation of conditions that support exploration, adaptation, and learning. Importantly, the case studies demonstrate both opportunities and tensions related to evaluating the extent of accessibility, which may be understood as a matter of an experiential spectrum, shaped by the structures available for exploration, the flexibility of interaction, and opportunities for developing transferable skills over time. While accessible data visualisation represents only one application domain, the broader implications of Flow-Centred Design may extend across accessibility research more widely, including education, assistive technologies, and access to physical or hybrid environ-

ments.

4 DISCUSSION

Flow has been widely applied in game design, educational technologies, creativity support systems, virtual and augmented reality, and user experience research as a framework for understanding engagement, immersion, motivation, and skill development. For example, game design has extensively adopted flow as a framework for balancing challenge and skill, structuring progression, and maintaining player engagement alongside broader usability and interaction principles[5]. Related ideas have also influenced areas such as educational technology, persuasive systems, and user experience design, where designers seek to sustain motivation, immersion, and intrinsic reward over time. In this sense, FCD does not introduce flow into HCI for the first time, but repositions it as a lens for accessibility evaluation, arguing that the experiential quality of accessible interaction deserves equal consideration alongside technical access and usability.

Importantly, Flow-Centred Design is not intended to replace established approaches such as User-Centred Design (UCD) or Human-Centred Design (HCD). Rather, FCD extends existing design frameworks by introducing flow as an additional evaluative lens for accessibility within the interaction design cycle. UCD and HCD have made substantial contributions to improving usability, reducing friction, and aligning technologies with human needs and contexts of use. However, accessible interaction is not solely determined by whether users can complete tasks or navigate interfaces, but also by the experiential quality of participation. Just as HCD expanded the scope of evaluation beyond usability toward broader user experience[15], FCD expands accessibility evaluation beyond access and usability — toward experience design and developmental flourishing[20]. FCD therefore complements existing approaches by asking not only whether interactions are usable and inclusive, but whether they support sustained engagement, reversible and low-stakes exploration, as well as long-term skill development in meaningful activities.

Emerging technologies create new opportunities for operationalising this approach. HCI has long recognised that no single interface can accommodate the diverse needs, preferences, and abilities of all users[18], motivating earlier movements in customisation, personalisation, end-user development, and malleable software[9, 13]. While such approaches have historically been constrained by implementation complexity and commercial incentives favouring standardised experiences, advances in artificial intelligence, multimodal interaction, and adaptive systems make personalised accessibility increasingly feasible. AI-assisted interfaces can dynamically adjust levels of guidance, feedback, modality, and task complexity in response to users' evolving skills, goals, and contexts of use. Although these capabilities must be implemented carefully to avoid over-automation, unpredictability, or diminished user agency, they offer promising mechanisms for maintaining an appropriate balance between challenge and skill over time — a central condition for enabling flow in accessible interaction.

4.1 Limitations and future work

Rather than proposing a definitive measurement framework, this paper offers an initial set of candidate measures grounded in the design principles of Flow-Centred Design. These measures are intended to

illustrate how the experiential quality of accessibility may be operationalised and evaluated in practice, while inviting further empirical refinement by the HCI and accessibility research communities. Future work may establish more consistent, valid, and robust approaches for evaluating the extent to which interactive systems support conditions for flow in accessible experiences.

A natural next step is methodological: empirically investigating how the design principles of Flow-Centred Design may be systematically observed and measured across different interaction contexts. For example, future work may identify behavioural indicators, qualitative accounts, and longitudinal patterns associated with *sustained engagement*, *guided exploration*, and *skill development*. In the context of *sustained engagement*, indicators may include reduced interruptions to concentration, fewer corrective actions, lower reliance on compensatory strategies, or qualitative reports of immersion, reduced fatigue, and intrinsic reward. For *guided exploration*, future studies may examine orientation, confidence, and contextual awareness during navigation, including the reversibility and predictability of interaction pathways, alongside qualitative accounts of perceived control and disorientation. Finally, *skill development* may be evaluated through changes in proficiency over time, such as reduced support needs, transferable interaction strategies across tasks, and increased confidence or autonomy in meaningful activities.

Such efforts would enable Flow-Centred Design to progress from a conceptual framework toward a more systematic methodology for evaluating accessibility as an experiential quality. Importantly, this work should remain sensitive to the contextual and subjective nature of flow experiences, recognising that meaningful accessibility may manifest differently across users, assistive technologies, and domains of activity.

5 CONCLUSION

Accessibility solutions have made digital inclusion and social participation for blind and other disabled people possible, but possibility alone is not enough. A system may technically satisfy accessibility requirements while still imposing significant friction, fragmented navigation, or barriers to developing proficiency over time. This paper argues that accessibility evaluation should extend beyond whether interaction is possible, to consider the experiential quality of interaction and the conditions that support meaningful engagement.

To address this gap, I proposed Flow-Centred Design as a conceptual framework for specifying and evaluating accessibility through the lens of flow. Rather than treating accessibility as binary compliance, FCD reframes it as a matter of an experiential spectrum, concerned with the extent to which interactions support sustained engagement, guided exploration, and skill development. In doing so, the paper introduced families of measures — including customisability and adaptability, exploratory structure and navigation, and learning and transferable skill development — as provisional ways of evaluating the experiential quality of accessibility.

Case studies in accessible data visualisation further suggest that these ideas are not purely speculative. Emerging interaction techniques increasingly support richer forms of exploration, personalisation, transferable interaction structures, and multimodal engagement that resonate closely with the aims of Flow-Centred Design. While the proposed measures remain intentionally provisional and require further empirical validation, they offer a starting point for future work on evaluating the extent of accessibility beyond access alone.

ACKNOWLEDGEMENTS

I would like to thank Jonathan Zong, who inspired me to express my scholarly opinion on Flow-Centred Design in a position paper, as a result of reading his excellent work on citational justice. I'm also grateful for everyone who shared their feedback, including Marianna Obrist, Emily Yates, and Jonathan Zong. Last but not least, my gratitude goes to Ronald Grau, who encouraged me to submit my own contribution to the symposium on Artificial Intelligence and Cognitive Science in Education, at AISB 2026.

REFERENCES

- [1] Colin Barnes and Geoff Mercer, *Disability*, Polity Press, Cambridge, 2003.
- [2] Len Barton, *Disability, Politics and the Struggle for Change*, David Fulton Publishers, London, 2001.
- [3] Matthew Blanco, Jonathan Zong, and Arvind Satyanarayan, 'Olli: An Extensible Visualization Library for Screen Reader Accessibility', in *IEEE VIS Posters*, (2022).
- [4] James I. Charlton, *Nothing About Us Without Us: Disability Oppression and Empowerment*, University of California Press, Berkeley, 1998.
- [5] Jenova Chen, 'Flow in games (and everything else)', *Commun. ACM*, **50**(4), 31–34, (April 2007).
- [6] Mihaly Csikszentmihalyi, *Finding Flow: The Psychology of Engagement with Everyday Life*, Basic Books, 1998.
- [7] Mihaly Csikszentmihalyi, *Flow: The Psychology of Optimal Experience*, Harper & Row, 2009.
- [8] Frank Elavsky, Lucas Nadolskis, and Dominik Moritz, 'Data Navigator: An Accessibility-Centered Data Navigation Toolkit', *IEEE Transactions on Visualization and Computer Graphics*, 1–11, (2023).
- [9] Gerhard Fischer, 'Meta-design: Beyond user-centered and participatory design', in *HCI International*, (2003).
- [10] Giles Hamilton-Fletcher, Marianna Obrist, Phil Watten, Michele Mengucci, and Jamie Ward, "'i always wanted to see the night sky': Blind user preferences for sensory substitution devices", in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI '16, p. 2162–2174, New York, NY, USA, (2016). Association for Computing Machinery.
- [11] Shuli Jones, Isabella Pedraza Piñeros, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan, "'Customization is Key": Reconfigurable Textual Tokens for Accessible Data Visualizations', in *Proceedings of the CHI Conference on Human Factors in Computing Systems* (CHI '24), p. 14, Honolulu, HI, USA, (2024). ACM.
- [12] Jonathan Lazar, Aaron Allen, Jason Kleinman, and Chris Malarkey, 'What frustrates screen reader users on the web: A study of 100 blind users', *Int. J. Hum. Comput. Interaction*, **22**, 247–269, (05 2007).
- [13] Henry Lieberman, Fabio Paternò, Marcus Klann, and Volker Wulf, 'End-user development: An emerging paradigm', in *End User Development*, 1–8, Springer, (2006).
- [14] Catherine Mei, Josh Pollock, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan, 'Benthic: Perceptually congruent structures for accessible charts and diagrams', in *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '25, New York, NY, USA, (2025). Association for Computing Machinery.
- [15] Donald A. Norman, *The Design of Everyday Things: Revised and Expanded Edition*, Basic Books, New York, 2013.
- [16] C. Norsworthy, B. Jackson, J. A. Dimmock, and B. T. Johnson, 'Advancing our understanding of psychological flow: A scoping review of conceptualizations, measurements, and applications', *Psychological Bulletin*, **147**(8), 806–827, (2021).
- [17] Richard K. Scotch, *From Good Will to Civil Rights: Transforming Federal Disability Policy*, Temple University Press, Philadelphia, 1984.
- [18] Ben Shneiderman and Pattie Maes, 'Direct manipulation vs. interface agents', *Interactions*, **4**(6), 42–61, (November 1997).
- [19] C. R. Snyder and S. J. Lopez, *The Oxford handbook of positive psychology*, Oxford University Press, 2011.
- [20] Yuanyang (YY) Teng and Darren Gergle, 'Access is not enough: Toward developmental flourishing', in *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing*

Systems, CHI EA '26, New York, NY, USA, (2026). Association for Computing Machinery.

- [21] Lily M. Turkstra, Tanya Bhatia, Alexa Van Os, and Michael Beyeler. Assistive technology use in domestic activities by people who are blind, 2025.
- [22] W3C. Web accessibility evaluation tools list.
- [23] Jonathan Zong, Crystal Lee, Alan Lundgard, Jaemin Jang, Daniel Hajas, and Arvind Satyanarayan, 'Rich screen reader experiences for accessible data visualization', (2022).
- [24] Jonathan Zong, Isabella Pedraza Piñeros, Ming Chen, Daniel Hajas, and Arvind Satyanarayan, 'Umwelt: Accessible structured editing of multimodal data representations', (2024).
- [25] Jonathan Zong, Isabella Pedraza Pineros, Mengzhu Katie Chen, Daniel Hajas, and Arvind Satyanarayan. Semantic scaffolding: Augmenting textual structures with domain-specific groupings for accessible data exploration, 2025.